# Daniel Charles McShan, PhD

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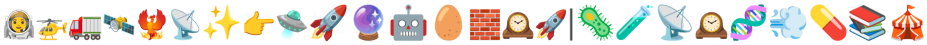

Principal Engineer, 5/5/2000-, Syzygyx

- 🥚 Naphome AI Sleep Deivce Engineering, *ESP-IDF, Gemini STT/LLM/TTS, Acoustics, REW, ESP32-S3, RK3308B*
- 🤖 Quadruped HiWonder PuppyPi Engineering, *OpenAI, ElevenLabs TTS/STT, AudD, Piper TTS, Whisper STT*
- 💨 Electronics Engineering support for a Tuya based Smart Air Purifier for deployment in India, *Tuya*
- 🚗 Design and development of embedded AI for a kids car toy
- 🤖 Humanoid Robot Engineering, *Unitree G1, Blender, BVH, MuJoCo, NVIDIA Isaac Sim, ROS2, DDS*
- 👷 Simulation and Test Architect, **NASA Orion I** used Rhapsody, Kennedy-Carter and MagicDraw *UML* and *SysML* tools to design a real time simulation architecture leveraging *Simulink GNC models* with *NASA Trick* and *JEOD* orbital dynamics using *Object Oriented Design* and C/C++ implementationon a Time Triggered Gigabit Ethernet (TTGbE) flight bus, for ar *DO-178B-eaque human rated mission while navigating a international partnership between Honeywell (C&DH), TTTEch (TTGbe) USA (sim sw) and other suppliers.*
- 🚁 Systems Engineering Manager, Drone Control System Project *Simulink, Qt, PixHawk, Ground Control System, Simulation, Avionics, novel ducted fan multicopter*
- 🚚 Systems Engineering Manager, Vehicle Telematics Project *Arduino, C++, Android, iOS, PCB Design, GPS, Cellular, OBD-II, CAN, Agile, JIRA*
- 🛰️ Ground Software Manager, **GeoEye-2** *C++, EGSE, Python, MS Project, vxWorks, RAD750 for a terrestrial imaging satellite*
- 🦞 Lead Software Engineer, **Mars Phoenix** Landing Radar Testset *EGSE, EGI, vxWorks, C++, MIL-STD-1553, Helicopter drop tests at Edwards AFB*
- 🛰️ Realtime simulation development support for **Mars Reconnaissance Orbiter** payloads *C++, Simulation, Rhapsody, UML*
- 📡 Developed realtime architecture for the tri-service **Multi Service Target Control System** *C++, Realtime Linux, Realtime Executive*
- ⌖ Distributor & Integrator, Hexamite Ultrasonic Positioning Systems

Articles

McShan, D.C.. "Realtime Simulation For Space Systems." *ISDC 2001: The National Space Society's 20th Annual International Space Development Conference*. 2001.
McShan, D. C. "Object oriented approaches to faster, better, cheaper testing." *19th Aerospace Testing Seminar, Manhattan Beach, CA*. 2000.
McShan, D.C., "How to Crash a Spacecraft" NASA Design for Safety, NASA Ames, 2000

Proposals

'Spacecraft Component Library", NASA SBIR 2001
"Fellowship of the R.I.N.G, Robotic Infrastruxture for Nomadic Groups", NASA Institute for Advanced Concepts 2001
"xORB: Aggregation of NASA center GNC software into a MATLAB Simulink graphical model based environment" NASA SBIR 2015
"DIMA-X, Distributed Integrated Modular Avionics for Exploration", NASA SBIR 2015

Methods (Infinity Incorporated - Intergalactic realm of eternal black darkness, white darkness)

- 🛸 Multiplicity adjustment, readjustment synthesis
- ⚗️ Isotope teleportation, transport liquidation, frequency polarization

Custodian, 2025-, Inquiry.Institute

🕯️ Developing an AI University run entirely by AI. Our faculty are AI agents of historical figures. Imagine learning history from Churchill or quantum physics from Einstein...

CTO, 2025-, Formul8.ai,

🌿 Developed ann AI-powered Marijuana Platform for Global Solutios, a comprehensive AI-powered consultation platform for the cannabis industry featuring 9 specialized AI agents with multi-agent orchestration capabilities. The platform provides intelligent workflow optimization through RAG-enhanced agents covering compliance, formulation (with RDKit molecular analysis), operations, sourcing, marketing, scientific research (PubMed integration), spectral analysis, patent/trademark research, and customer success. Built with TypeScript/React frontend, Node.js/Express backend, PostgreSQL database, and integrated GitHub repository management for automated issue tracking and feature development across all agent repositories.

CoFounder, 2019-, Terpedia

🌷 Terpedia is the encyclopedia of Terpenes

Created a knowledge base in *OpenLink Virtuoso*, a multimode DBMS that integrates relational databases as well as knowledge graphs (CheBI, RHEA) in OWL to capture knowledge and relationships about Terpenes. Terpedia uses an LLM front end and to understand user requests and answer with both LLM knowledge and knowledge from the DBMS.

VP of Engineering, 2020-2025, Advanced Vape Supply

💨 Software development, PCB development, Molecule Control Tower smart vape mod. Uses an array of sensors for precision temperature control of an Energized Silicon Carbide Crucible. Developed a PID controller for a Pulse Skip Modulation (PSM) boost circuit. *STM32, ESP32, C, C++, Micropython, LVGL, Thermal Modelling,*

Principal Engineer, 2023-, HyperLocal.Global

✨ Research & Development of sustainable, local AI systems for hyoerlocal weather, horticulture, security, home control, autonomous garden vehicles, communication and control, tactical awareness. *TensorFlow, Nano Jetson, Tesla P4, Google Coral TPU, TAK, LLAMA 3.1*

Founder, 2011-2012, 2024- Xybotyx, LLC

🤖Managed and developed corporate strategy, design and development of a consumer iPhone robot. Xybot 1.0 was programmed in AVR C/C++ and C# on the iPhone
🤖Son of Xybot - coming soon - leverages a WaveShare ROVER with ESP32 and *Swift, iOS ARKit, TensorFlow* on the iPhone. It's first mission will be used to scan and analyze the desert ground looking for Matter Out Of Place (MOOP)

Founder, 2024-, Fortun.ai, *Sufficiently Advanced Technology*

🔮Personal AI fortune teller, uses a custom GPT-4 LLM to input runes, tarot, etc. along with a query and returns a detailed individualized fortune based on knowledge of the user over time and from a private profile. *RFID, NFC, ESP32, Wordpress, PHP, ChatGPT API*

Principal Engineer, 2017-2018, Geneva Technologies

📡Principal engineer on a SBIR Phase II for modernizing a F-16 RADAR *Compact Antenna Test Range* lab at Hill Air Force Base.

Founder, 2013-2014, BRYXco

🧱Manage and developed Arduino programmable, iPhone controllable, LEGO compatible brick *Arduino, C++, 3D printing, PCB design and development*

Embedded Systems Engineer, 1996-2000, Lockheed Martin, Littleton, CO

🕰️Real Time Simulation Lab: Pioneer of Linux and C++ for Realtime Simulation. Installed the fist Linux system in LMCO. Ran the first C+= on vxWorks.
🛰️Astronautics: Designed, developed, integrated, tested and maintained the real time hardware in the loop testbed for **Mars 98 Orbite**r, **Mars 98 Lander**, **Stardust**, **Spitzer**, **Genesis**, **Mars 01**

Embedded Software Engineer, 1994-1996, FAAC, Inc., Ann Arbor, MI

🚀Developed and integrated faster than real time weapon simulations for F-15 AGM-130 and GBU-15 Zone Acquisition Process *DOD-STD-2167A, Ada*

Chief Science Officer, 2019-2023, GoodFOR

💊Formulate and develop diverse oral and topical pharmaceuticals and cosmetics, including FDA OTC Drugs

Founder, 2020-, InCivi.co

📈Conceived and developed an eCommerce system for inexpensive citizen science clinical trials. *WordPress, PHP*

Founder, 2019-, Apothyx

💧Developed concept and demo technology for the Robotic Apothecary based on OpenTrons OT-2 platform, *Python*

Staff Scientist, Software Engineering, 2017 Muse Biotech

🧬Designed and developed embedded control software for Inscripta Onyx automated CRISPR platform. *Python*

Founder, 2010-, The Exzyme Corporation

🦠Developed my dissertation research for retrobiosynthesis - a method for finding biosynthetic pathways based on E.J.Corey's Nobel Prize research.

Proposals

"ExoBioME: Exobiological Metabolic Engineering", NASA Institute for Advanced Concepts 2013

## Academic

🧠Postdoctoral Fellow, Dept. of Pharmacology, 2016-17, University of Colorado School of Medicine, Aurora, CO

Developed metabolic knowledge base and biochemical informatics system *SPARQL, Clojure, AllegroGraph, Java, JChem*

🧠Ph.D., Computational Bioscience, 2012, University of Colorado School of Medicine, Aurora, CO

Dissertation: *"Optimal Search Algorithms for the Analysis and Synthesis of Metabolic Systems" Allegro Common Lisp*

Patents

Methods, Systems and Software for Predicting Metabolic Pathways

Articles

I Shah, D McShan, Methods, Systems and Software for Predicting Biochemical Networks or Pathways WO Patent 2,005,054,978
DC McShan, Towards inference of a biochemical ontology from a metabolic database. Comparative and functional genomics 6 (7-8), 398-406 2005
D McShan, I Shah, Heurstic search for metabolic engineering: de novo synthesis of vanillin, Computers & chemical engineering 29 (3), 499-507, 2005
DC McShan, M Updadhayaya, I Shah, Symbolic inference of xenobiotic metabolism. Pacific Symposium on Biocomputing, 545, 2004
DC McShan, S Rao, I Shah, PathMiner: predicting metabolic pathways by heuristic search. Bioinformatics 19 (13), 1692-1698, 2003

🧠B.S.E. (Electrical), 1994, University of Michigan College of Engineering, Ann Arbor, MI

## Other

CEO, Founder, Brûlée,co , 2013-

🎪501(c)3 Organization for the Arts

Manage and Operate a 200 person Burning Man Esplanade Circus/Carnival/Cabaret Theme Camp